AYUSH JUVEKAR

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IoT / Embedded Engineer

PROFILE

IoT and embedded-systems engineer focused on Arduino/C++ sensor prototypes, MQTT/Zigbee communication, Node-RED automation, and edge-device demos.

EDUCATION

Michigan Technological University, Houghton, MI — **CGPA: 3.9/4.0**
M.S. Computer Science — Graduated Apr 24, 2026

Pune Institute of Computer Technology, Pune, India — **GPA: 8.14/10**
B.E. Computer Engineering — Jun 2024

EXPERIENCE

Teaching Assistant - Physics Department | Michigan Technological University — Sep 2024 - Apr 24, 2026
Houghton, MI

- Supervised and graded physics labs for 100+ students across four semesters, guiding experiments, explaining concepts, and providing clear feedback across five lab sections per semester.

SELECTED PROJECTS

IoT Air Quality Monitoring System | *Arduino Uno, C++, MQTT, Zigbee, Node-RED* | IEEE

- Co-built an IEEE-published IoT air-quality prototype for CO/gas monitoring, using Arduino Uno/C++, MQTT/Zigbee communication, and Node-RED flows to automate air-filter behavior from live sensor readings.
- Owned the Node-RED automation and conference presentation for the two-person project, contributing to the paper/report accepted at ESCI 2023.

Crowd Anomaly Detection Prototype | *Python, TensorFlow, OpenCV, TensorFlow Lite, Jetson Nano*

- Led model research and data collection for a BE final-year team project on edge crowd-anomaly detection, recording training videos and evaluating CNN/TensorFlow approaches on 10K+ frames.
- Optimized the prototype with TensorFlow Lite for Jetson Nano demonstration, reaching about 92% test accuracy in controlled demo runs without claiming production deployment.

Particle Vibration Analysis | *MATLAB, Computer Vision, Object Detection, ZED 2 Camera*

- Built MATLAB computer-vision scripts for particle vibration analysis from ZED 2 camera video, using object detection workflows to track particles in test recordings.
- Delivered analysis outputs for a professor's downstream PhD research, validating scripts on sample particle videos before larger research use.

TECHNICAL SKILLS

Languages	C++, Python, MATLAB
Embedded / IoT	Arduino Uno, sensor integration, MQTT, Zigbee
Automation	Node-RED flows, live sensor-triggered control
Edge / Computer Vision	Jetson Nano, TensorFlow Lite, ZED 2 camera video
Data / Analysis	Jupyter Notebook, NumPy, Pandas
Tools	Git, Linux

PUBLICATION

A. Juvekar et al., “Carbon Monoxide Concentration Monitoring System,” *ESCI 2023*, presented Mar 2023. IEEE